

Metro Signage HMI — Screen Functions

A page-by-page description of the operator touch panel: what each screen is for, what it shows, and the control rules it must obey.

Version **1.0** September **2026** Screens **5** Panel **1280 × 800**

The HMI is a **local touch panel** that monitors and manually overrides up to **100 ESP32 signage nodes** across a station. It is a viewer and command surface only — the gateway daemon remains the master, so an HMI restart never interrupts the live network. Each node drives a signboard with a **left moving arrow**, a **right moving arrow**, and two always-on static zones. The screens below share one status language, applied consistently everywhere.

STATUS COLOUR LANGUAGE — USED ON EVERY SCREEN

- | | |
|--|--|
|  Green — Healthy. Online and operating as commanded. |  Amber — Warning. Attention needed, still operating. |
|  Red — Fault. A failure that needs action. |  Grey — Offline / unknown. Not reporting. |

01 Fleet Overview

Home screen · whole-network status at a glance

PURPOSE

The landing screen. It answers one question instantly — **is the fleet healthy, and if not, what needs attention?** — without the operator opening a single node.

ON SCREEN

- **Status grid.** A 10 × 10 grid of tiles, one per node, each colour-coded by health. Green tiles mean everything is operating correctly; amber, red and grey flag the exceptions. A tile is tappable to open that node.
- **Headline counts.** A large “online out of 100” figure, with secondary counts of nodes that are healthy, in warning, in fault, and offline.
- **Alarms rail.** A running list, down the right, of conditions that need a person — newest first, with a count badge. Each entry shows the node, the cause, and a timestamp, and taps through to the node.

WHAT RAISES AN ALARM

The rail is built from real conditions the system already detects: a sensor reporting FAIL_OPEN or FAIL_SHORT; a node whose commanded mode and actual mode disagree; a sign left latched in a technician mode; and a node that has gone offline or cannot verify its outputs.

02 Signage List

Searchable index · global control source

PURPOSE

The directory of all 100 nodes, for finding a specific sign and for switching the whole fleet between automatic and manual control.

ON SCREEN

- **Node table.** Each row lists the node ID, IP address, assigned holding register, and a single roll-up health indicator. Rows are searchable by ID or IP and open the node when tapped.
- **Global control-source toggle.** A prominent switch between **REMOTE (SCADA / automatic)** and **LOCAL (HMI / manual)**. Its state is unmistakable, and it is deliberately hard to flip by accident.

ROLL-UP STATUS

INDICATOR	MEANING
● OK	Online, no faults, running its commanded mode.
● Warning	Latched in a technician mode, or commanded ≠ actual.
● Fault	A sensor failure, a hardware fault, or unreachable.
● Offline	Not reporting to the gateway.

THE CONTROL SOURCE IS GLOBAL

The toggle writes a single system register that governs **all 100 nodes at once** — it is not per-node. When set to REMOTE, the manual mode buttons on every Node Detail screen are locked; when set to LOCAL, they become live. This is the only control-source switch in the system; there is no physical local/remote switch anywhere.

PURPOSE

Everything about a single sign, split into what it is **reporting** (read-only telemetry) and what the operator can **command** (arrow control). Opened by tapping a node anywhere else in the app.

TELEMETRY

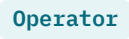
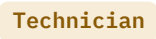
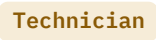
Live rows for connection, main power, the two arrows, the two static zones, and a backup battery. The four arrow and zone rows use a four-state reading — and it is essential that an intended-off output is **not** shown as a fault:

STATE	COLOUR	MEANING
ON	 Green	Intended on, current flowing — healthy.
OFF	 Neutral	Intended off, no current — also healthy .
FAIL_OPEN	 Red	Intended on but drawing no current — broken LED, wire, or driver.
FAIL_SHORT	 Red	Intended off but drawing current — a short or welded driver.

The backup battery is shown permanently greyed as “N/A · Not fitted”; the hardware slot exists but no battery is reported yet.

ARROW TRIGGERING

The core control. Commands drive the **moving arrows only** — the static zones are hardwired always-on and are never switchable. Two tiers of mode:

VALUE	MODE	ACCESS
0 / 1 / 2	Arrows off · LHS chase · RHS chase	 everyday use
3	Both chase	 latches
4 / 5 / 6	Solid on — left · right · all	 latches

COMMANDED VS ACTUAL

The node reports what it is **actually executing**, shown beside what it was **commanded**. When the two agree the panel reads “in sync”; when they diverge it raises a prominent mismatch warning — the one case where a sign is showing something other than what the system believes.

TECHNICIAN MODES LATCH – BY DESIGN

Modes 3–6 have no auto-revert: a sign left solid-on stays lit until someone changes it. The interface therefore styles them as a caution state and surfaces them in the fleet alarms, so a latched sign is never lost among 100 nodes. The mode buttons are live only when the global control source is LOCAL.

04 Per-LED Control

Modal over Node Detail · low-level override

PURPOSE

A technician-level pop-up for driving the ten arrow LEDs individually, kept out of the everyday flow because it is not normal operation.

ON SCREEN

- **Ten LEDs.** The left and right arrows as five lamps each. Tap a lamp to turn that LED on or off; lit lamps glow in the accent colour.
- **Quick actions.** “All LEDs on” and “All off” shortcuts.
- **Live raw value.** The resulting command value updates as lamps change, so the exact pattern being sent is always visible.

ENCODING & SAFETY

Per-LED patterns are sent as a raw value from 10000 (all off) to 11023 (all on) — a 10000 base plus a ten-bit mask. Because these outputs also latch, the pop-up carries a standing caution that the sign stays in whatever pattern is applied until a normal mode is re-commanded. In the current developer build the pop-up opens without a PIN; a technician PIN gate is planned.

05 Diagnostics

System health · maintenance action

PURPOSE

The health of the gateway and the network behind the panel, plus a single maintenance action.

ON SCREEN

- **Service health.** Live status of the MQTT broker, the Modbus TCP server, the gateway service, and the master gateway IP.
- **Network counts.** Large online / offline figures, with secondary warning and fault counts.
- **Broadcast ping.** One button that asks every node to re-publish its full telemetry.

THE PING IS A REFRESH, NOT A FAIL-SAFE

Broadcast ping only requests fresh telemetry so the panel can rebuild state quickly. It does **not** reset any safety timer — none exists. On network loss, nodes hold their last command; pressing ping more often does not make the system safer, and it must never be described to an operator as a safety mechanism.